

MyDesign Portfolio

Element K

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- defining the problem (elements A-C)
- ideation (element D)
- selection & prototype design (elements D-F)
- prototype building (element G)
- testing (elements H-I)
- evaluation (element J)

For each project step, explain: What went well? What didn't go well? Why?

Lessons Learned About the Engineering Design Process (K3)

What do you think designers need to consider or do based on your experience? A level 5 reflection illustrates thorough and specific details on improving the design process in the future. Make sure to touch on lessons from across the whole engineering design process and experience that can apply to any future engineer. The lessons learned in Element K should focus on the engineering design process and how you implemented it. Later on (in L1) you will provide recommendations for continuing work on this project in particular.

Summarize each step of the design process including:

Engineering Design Process: Fixed Cable Trough Project

Initial Meeting in TV Studio:

The project began with a meeting in the TV studio, where Mr. O'Farrell introduced several potential projects needing solutions. Our group chose to work on the cable trough because we recognized the urgency and importance of protecting the studio's cables. The current situation—with exposed, tangled wires—posed both a safety hazard and a risk of technical failure. We knew we could design a solution that provided a clear, walkable path while protecting the cables from damage.

Creation of a Design Brief:

Based on Mr. O'Farrell's outlined needs, we created a design brief for a fixed cable management system. We researched cable troughs, defined the necessary functions—like protecting cables and allowing access to the rear of the computer table—and set design constraints such as the dimensions (236" L x 18" W x 3" H). The brief also established our design priorities: durability, functionality, and efficient cable organization.

A-C: Defining the Problem:

In Element A, we clearly defined the problem: disorganized cables in the TV studio that are frequently stepped on, creating safety risks and potential loss of files if disconnected. We researched best practices in cable management and analyzed how a physical cover would resolve both safety and organizational concerns. This step in the engineering design process is probably one of the most important steps because without a clear definition of the project, outlining objectives, criteria and constraints a viable solution would be hard to achieve.

In Element B (documentation and analysis of prior design attempts), we researched what limited solutions existed and why they failed. We found little to no existing infrastructure addressing this issue, making the problem worse over time.

In Element C (presentation and justification of solution design requirements), we established our design requirements: fixed installation (no need to move cables), walkable surface, protection from pressure, and adherence to the studio's specific spatial

dimensions. These priorities came directly from user feedback and studio needs. We went through the design requirement and ranked them either as high or low priority.

D: Prototype Designing and Ideation:

In Element D, we brainstormed and sketched several cable trough models. We created 3 concept sketches, evaluated them against our design requirements, and selected the best one to move forward with. Using a design decision matrix helped us with the different requirements and decide which idea to go with. Using Onshape, we developed CAD models and continually refined them. This stage also included creating and maintaining a Gantt chart to manage our workflow, though staying on track with it was sometimes challenging.

Note: For this project we did not complete E and F which was application of STEM principles and making a case for how the chosen design would work.

G: Prototype Building: Element G was when we began turning our design into a reality. Using CAD, we created multiple assemblies of the cable trough using the school's Onshape website. We iterated four different prototypes, testing each for strength, fit, and function. It was also important for us to measure while we were going. Our initial wood concept failed because it was not see-through and it was brittle. So we revised the design to make an acrylic plexiglass panel which was transparent while also being durable.

H-I: Testing:

In Element H, we conducted rigorous testing. We evaluated the prototypes based on how well they protected cables, supported foot traffic, and maintained organization. We ran trials simulating real-world use and collected data, as well as expert feedback.

In Element I, we visualized our results and our tests showed that the final model met all our high priority requirements effectively, and feedback from others confirmed its practicality.

J: Evaluation:

In Element J, we presented our final product to stakeholders—including Mr. O'Farrell, the Walpole Media Outlet, TV production studio students. We also presented to field experts including Mr. O'Farrell, Professor. Love, Mr. Braccini, and faculty—during the Design-A-Thon. We gathered feedback and reviewed scores from rubrics. Our design received the highest marks in all categories besides Mr. Braccini who provided a 2/3 for identifying and describing a problem, usage of engineering, and articulating appropriate

STEM practices. We also documented the credentials and input from each stakeholder to validate the results.

K3: Lessons Learned About the Engineering Design Process:

This project showed us the value of a structured engineering design process. While early stages—like Elements A-I—felt repetitive, they were crucial to creating a well-thought-out and successful solution. Each phase built on the last, and testing proved especially important in refining our work. In the future, adding feedback from general users—not just experts—could make the design even more accessible. Overall, this process gave us a strong foundation for solving real-world problems thoughtfully and effectively.